

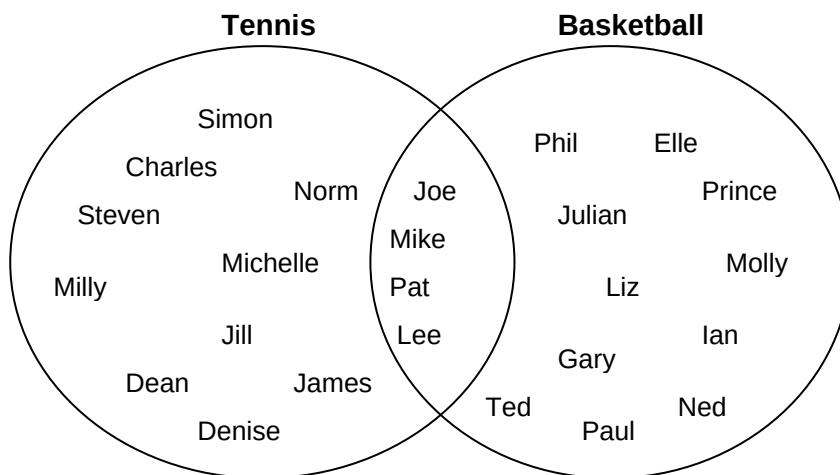


Probability in described events

Teacher resource

Question 1

The Venn diagram shows the names of students in a class who play tennis and those who play basketball.



What is the probability a student in the class plays basketball?

Answer: 60%, $\frac{3}{5}$ or 0.6

Skill: Students use data to calculate probability

Background information

Students should understand that probability is a measure of chance or how likely something is to happen. As a measure, the unit chosen is 1 to describe certainty and 0 to describe impossibility. Probability of any event therefore must be measured by a number between 0 and 1 inclusive.

For students to manage problems related to probability they must understand the concept of chance itself, the concept of measurement as well as the meaning of fractional numbers expressed as fractions, decimals, ratios and percentages. The difficulty of each of these concepts, coupled with their interaction in a chance situation, makes seemingly simple problems involving probability quite difficult for some secondary students.

Additional questions

1. What is the probability a student plays both tennis and basketball?
2. If a student plays tennis, what is the probability the student also plays basketball?

Question 2

Bella had 42 wrapped lollies. Some were caramels and some were mints. The wrapped lollies all looked the same.

Bella offered Jane a lolly. She had a 1 in 6 chance of taking a caramel.

How many mints did Bella have?

Answer: 35

Skill: Students use probability to make predictions.

Additional questions

1. What was the probability of Jane taking a mint?
2. Assume Jane took a mint. If Jane takes a second lolly, what is the probability that she takes another mint?

Question 3

David prepares a bag of counters so that 40% of the counters are blue. He removes one blue counter from the bag and does not replace it.

He then takes another counter from the bag without looking. What is the probability that this second counter is blue?

- ☐ less than 40%
- ☐ equal to 40%
- ☐ more than 40%

Answer: A

Skill: Students calculate probability.

Additional questions

1. What is the probability that the second counter is not blue?
2. A bag contains blue, red, green and yellow counters. There are equal proportions of each colour. A blue counter is removed. What is the chance that the second counter is yellow?

Question 4

The chance of having a female child is equal to the chance of having a male child.

A family has three children.

What is the probability that the 3-child family has 2 boys and 1 girl and that the eldest child is a male?

Answer: 25%

Skill: Students calculate probability of a compound event.

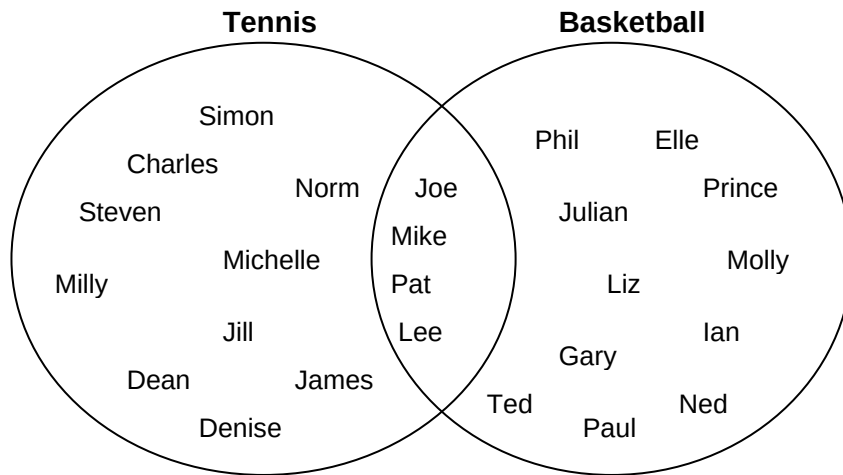
Additional questions

1. What is the probability that a 3-child family has more girls than boys?
2. What is the probability that a 3-child family has only boys **or** only girls?
3. Of the 3-child families that have more boys than girls, what is the probability that the eldest child is a boy?

Student activity: Probability in described events

Question 1

The Venn diagram shows the names of students in a class who play football and those who play basketball.



What is the probability a student in the class plays basketball?

Question 2

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Bella offered Jane a lolly. She had a 1 in 6 chance of taking a caramel.

How many mints did Bella have?

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